

Ex.no:9

MINI PROJECT – HEALTHCARE DATA ANALYTICS

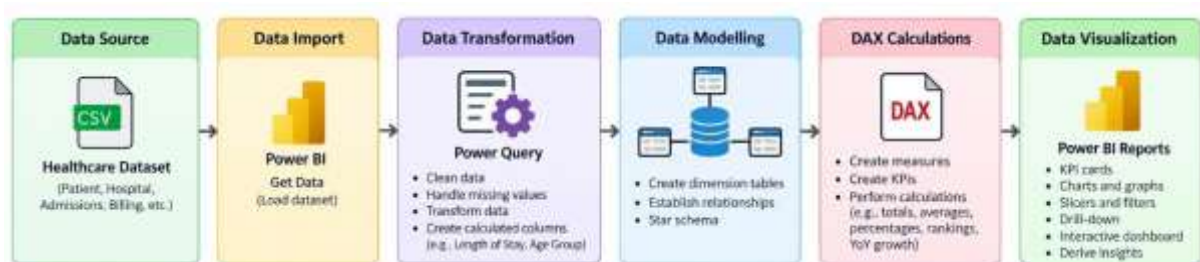
Aim:

To develop an interactive Healthcare Data Analytics Dashboard using Power BI to analyze patient information, medical conditions, hospital admissions, length of stay, billing, insurance providers, and test results.

Algorithm:

1. Import the Healthcare Dataset into Power BI and Tableau.
2. Clean and transform the data using Power Query in Power BI and Data Source in Tableau.
3. Check and correct the data types and remove unwanted or duplicate data.
4. Create required calculated fields such as Age Group and Length of Stay.
5. Create the required data model and relationships in Power BI.
6. Create required DAX measures and KPIs in Power BI.
7. Create calculated fields and required measures in Tableau.
8. Create charts for patients, medical conditions, admissions, hospitals, billing, insurance, and test results.
9. Add cards, KPIs, slicers, filters, and drill-downs in Power BI.
10. Add filters, dashboards, and interactive visualizations in Tableau.
11. Design and format the dashboards in both Power BI and Tableau.
12. Analyze the visualizations and identify important healthcare insights.
13. Present the final results through the interactive dashboards.
14. End.

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd
```

```
from sqlalchemy import create_engine, text

# Load dataset

df = pd.read_csv("healthcare_dataset.csv")

# Create a copy for processing

healthcare = df.copy()

# Remove duplicate records

healthcare.drop_duplicates(inplace=True)

# Handle missing values

healthcare["Name"] = healthcare["Name"].fillna("Unknown")

healthcare["Doctor"] = healthcare["Doctor"].fillna("Unknown")

healthcare["Hospital"] = healthcare["Hospital"].fillna("Unknown")

healthcare["Insurance Provider"] = healthcare["Insurance Provider"].fillna("Unknown")

healthcare["Medication"] = healthcare["Medication"].fillna("Unknown")

healthcare["Test Results"] = healthcare["Test Results"].fillna("Unknown")

# Convert date columns

healthcare["Date of Admission"] = pd.to_datetime(
    healthcare["Date of Admission"]
```

```
)
```

```
healthcare["Discharge Date"] = pd.to_datetime(
```

```
    healthcare["Discharge Date"]
```

```
)
```

```
# Create Length of Stay
```

```
healthcare["Length of Stay"] = (
```

```
    healthcare["Discharge Date"] -
```

```
    healthcare["Date of Admission"]
```

```
).dt.days
```

```
# Create Age Group
```

```
healthcare["Age Group"] = pd.cut(
```

```
    healthcare["Age"],
```

```
    bins=[0, 17, 30, 45, 60, 150],
```

```
    labels=["0-17", "18-30", "31-45", "46-60", "60+"]
```

```
)
```

```
# Standardize column names
```

```
healthcare.columns = (
```

```
    healthcare.columns
```

```
    .str.lower()
```

```
    .str.strip()
```

```
    .str.replace(" ", "_")
```

```
)
```

```
# Display original and cleaned shape
```

```
print("Original Shape:", df.shape)
```

```
print("Cleaned Shape:", healthcare.shape)
```

```
healthcare.head()
```

2. MySQL – Database and Data Loading

MySQL Command Prompt:

```
net start MySQL80
```

```
mysql -u root -p
```

MySQL:

```
CREATE DATABASE IF NOT EXISTS healthcare_analytics;
```

```
USE healthcare_analytics;
```

Python/Jupyter:

```
username = "root"
```

```
password = "YOUR_MYSQL_PASSWORD"
```

```
host = "localhost"
```

```
port = 3306
```

```
# Connect to MySQL database
```

```
db_engine = create_engine(  
    f"mysql+pymysql://{username}:{password}@{host}:{port}/healthcare_analytics"  
)
```

```
# Load cleaned data into MySQL
```

```
healthcare.to_sql(  

```

```

"healthcare_data",
con=db_engine,
if_exists="replace",
index=False,
chunksize=5000
)

print("Data loaded successfully into MySQL.")
Verify:
pd.read_sql(
    "SELECT COUNT(*) AS total_rows FROM healthcare_data",
    db_engine
)

```

3. Power BI – DAX Measures

Total Patients

Total Patients =

COUNTROWS(FactHealthcare)

Total Billing

Total Billing =

SUM(FactHealthcare[billing_amount])

Average Billing

Average Billing =

AVERAGE(FactHealthcare[billing_amount])

Average Age

Average Age =

AVERAGE(FactHealthcare[age])

Average Length of Stay

Average Length of Stay =

AVERAGE(FactHealthcare[length_of_stay])

Emergency Admissions

Emergency Admissions =

CALCULATE(

[Total Patients],
FactHealthcare[admission_type] = "Emergency"
)

Emergency Admission Rate

Emergency Admission Rate =

DIVIDE(
[Emergency Admissions],
[Total Patients]
)

Abnormal Tests

Abnormal Tests =

CALCULATE(
[Total Patients],
FactHealthcare[test_results] = "Abnormal"
)

Abnormal Test Rate

Abnormal Test Rate =

DIVIDE(
[Abnormal Tests],
[Total Patients]
)

Maximum Billing

Maximum Billing =

MAX(FactHealthcare[billing_amount])

Minimum Billing

Minimum Billing =

MIN(FactHealthcare[billing_amount])

Previous Year Billing

Previous Year Billing =

CALCULATE(
[Total Billing],
SAMEPERIODLASTYEAR(DimDate[Date]))

)

Billing YoY Growth

Billing YoY Growth % =

DIVIDE(

[Total Billing] - [Previous Year Billing],

[Previous Year Billing]

)

4. Tableau – Calculated Fields

If you also need the Tableau part in the same format, use these:

Total Patients

COUNT([Name])

Total Billing

SUM([Billing Amount])

Average Billing

AVG([Billing Amount])

Average Age

AVG([Age])

Average Length of Stay

AVG([Length of Stay])

Emergency Admissions

SUM(

IF [Admission Type] = "Emergency"

THEN 1

ELSE 0

END

)

Emergency Admission Rate

SUM(

IF [Admission Type] = "Emergency"

THEN 1

```
ELSE 0
END
)
/
COUNT([Name])
Abnormal Tests
SUM(
    IF [Test Results] = "Abnormal"
    THEN 1
    ELSE 0
    END
)
```

Abnormal Test Rate

```
SUM(
    IF [Test Results] = "Abnormal"
    THEN 1
    ELSE 0
    END
)
/
```

COUNT([Name])

Age Group

```
IF [Age] < 18 THEN "0–17"
```

```
ELSEIF [Age] <= 30 THEN "18–30"
```

```
ELSEIF [Age] <= 45 THEN "31–45"
```

```
ELSEIF [Age] <= 60 THEN "46–60"
```

```
ELSE "60+"
```


END

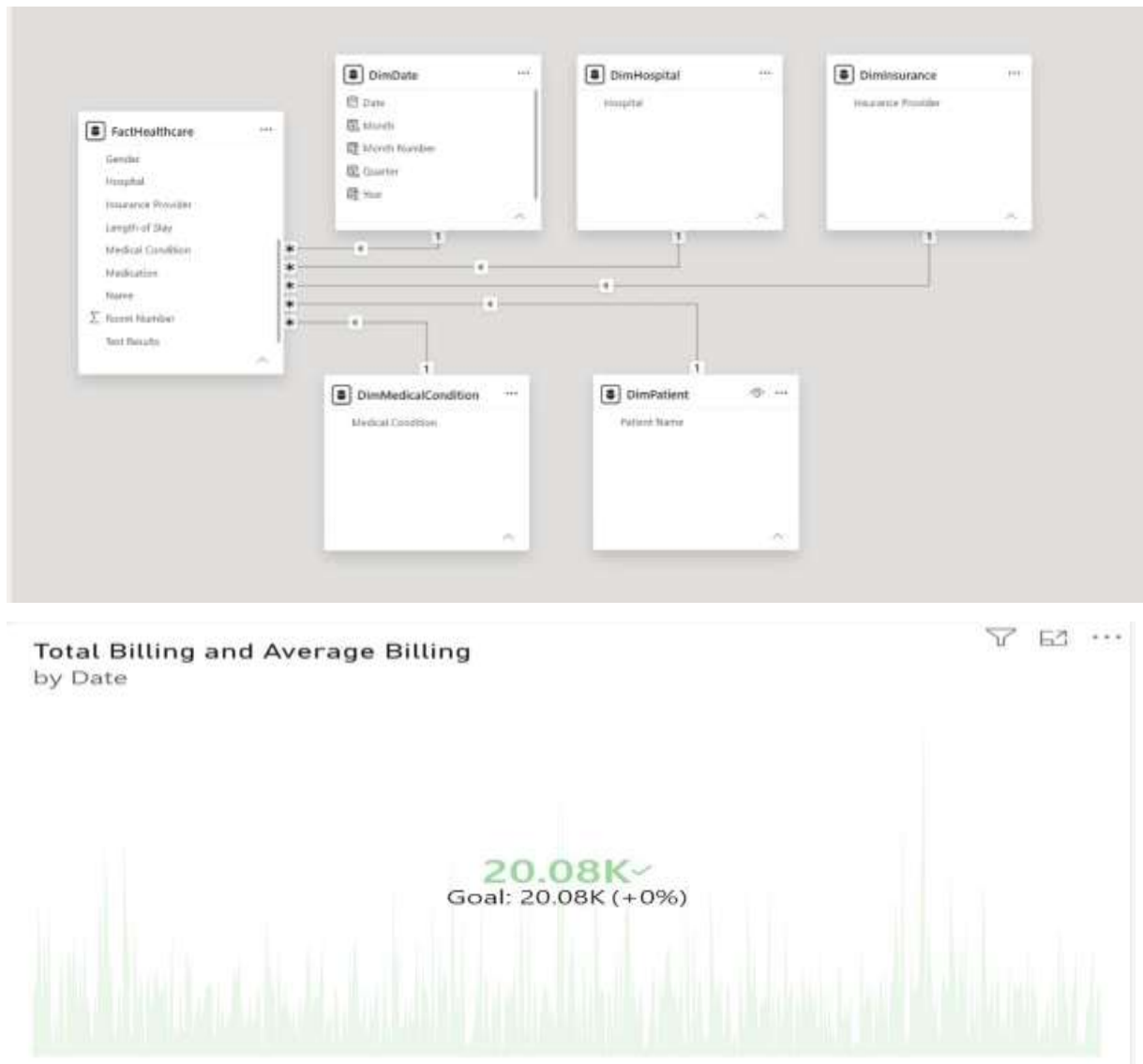
Length of Stay

```
DATEDIFF(  
    'day',  
    [Date of Admission],  
    [Discharge Date]  
)
```

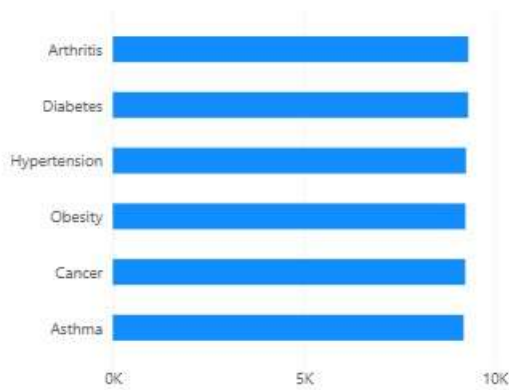
4. Tableau Dashboard

Created visuals in tableau in separate sheets and created a dashboard

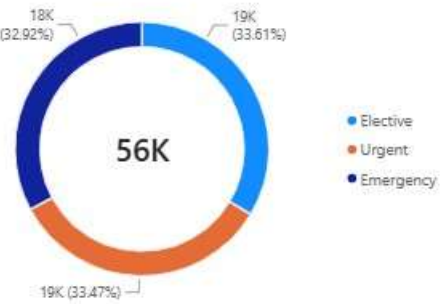
Output:



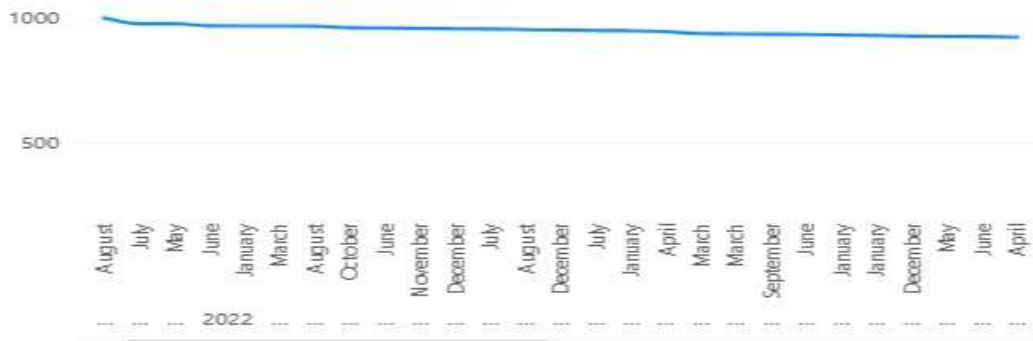
Total Patients
by Medical Condition



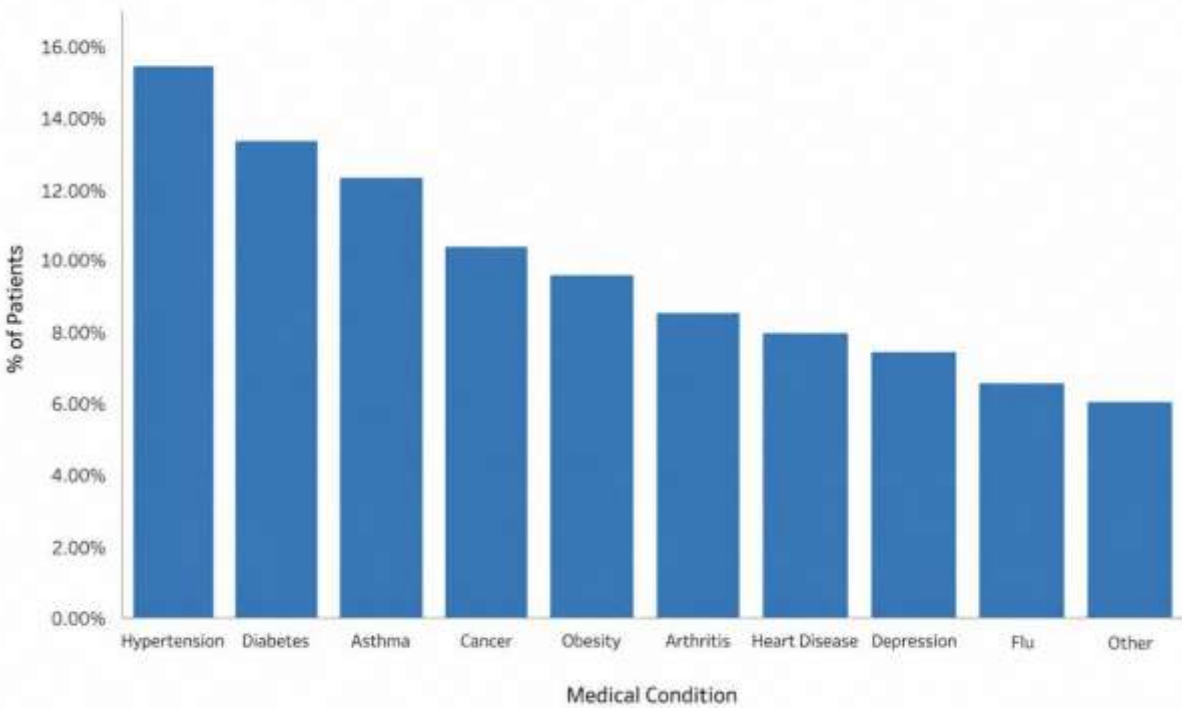
Total Patients
by Admission Type



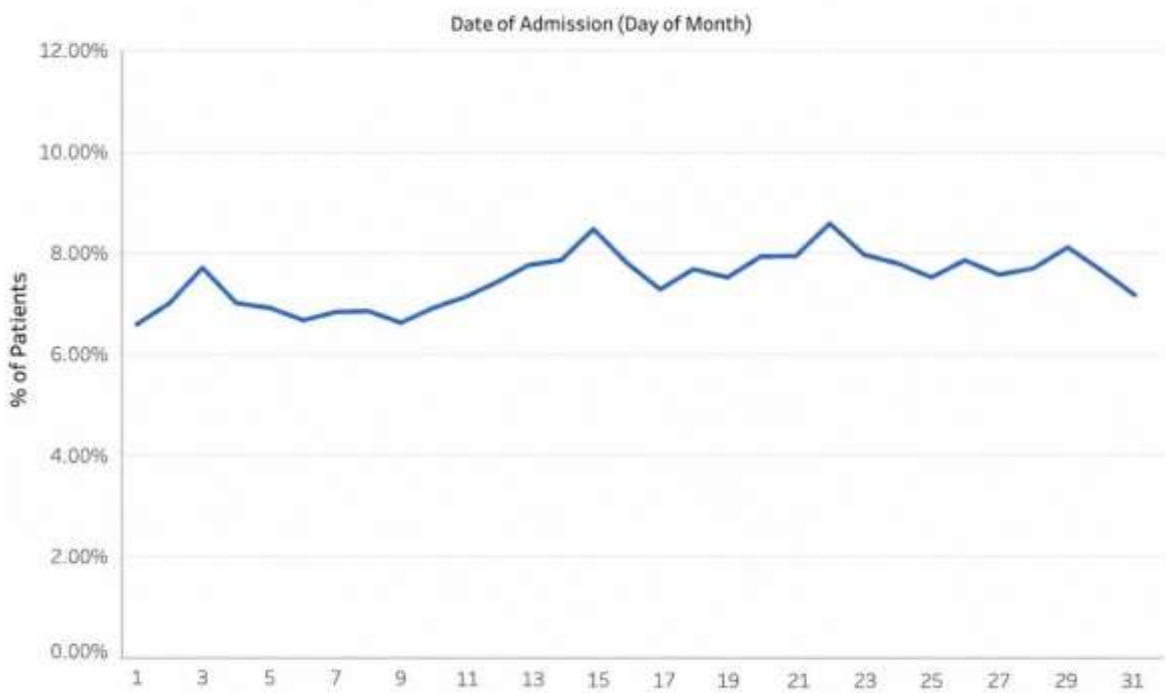
Total Patients
by Year and Month



Patients by Medical Condition



Patient Admission Trend





Result:

Thus, an interactive Healthcare Data Analytics Dashboard was successfully developed and designed using Power BI and Tableau. The healthcare dataset was processed using Python, stored in MySQL, and modeled into star schema dimensions to analyze critical patient metrics. Interactive KPIs, DAX measures, and visual dashboards were effectively created to display healthcare insights, including patient admission trends, medical conditions, length of stay, and billing analysis.